

## Technology Stack

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 26 June 2026                                                |
| <b>Team ID</b>       |                                                             |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 2 Marks                                                     |

### Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

### Technology Stack Details

| S.No | Architecture Component / Layer                          | Technology Chosen                | Justification / Purpose                                                                                                                                        |
|------|---------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side (HTML, CSS, Bootstrap)           | HTML5, CSS3, Bootstrap 5         | Provides responsive and mobile-friendly UI. Bootstrap ensures consistent layout and styling for the crop recommendation input form and results page.           |
| 2    | Backend / Server-Side (Python, Flask)                   | Python 3.10+, Flask              | Flask is lightweight and ideal for deploying ML models as web APIs. Python offers seamless integration with Scikit-learn and Pandas for data processing.       |
| 3    | Database / Data Storage (CSV / File-based)              | CSV Dataset + Pickle (.pkl)      | The agricultural dataset is stored as a CSV file. Trained ML models are serialized using Pickle for efficient reuse during crop prediction without retraining. |
| 4    | Cloud / Hosting / Deployment (Local / Flask Dev Server) | Flask Development Server (Local) | The application is deployed locally using the Flask built-in development server, allowing real-time crop recommendations through the browser interface.        |
| 5    | Version Control & CI/CD (GitHub, Git)                   | Git + GitHub                     | Git is used for version control and tracking code changes. GitHub hosts the repository for team collaboration, code review, and project submission.            |

| S.No | Architecture Component / Layer                                                    | Technology Chosen                                                  | Justification / Purpose                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6    | Third-Party APIs / Other Tools (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn) | NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook | NumPy and Pandas handle data processing. Scikit-learn provides ML algorithms (KNN, Logistic Regression, Random Forest, K-Means). Matplotlib and Seaborn are used for data visualization and analysis. |